AI & GENERATIVE AI TRACK / PROJECT 04 OF 05

# LearnMate

PHASE 05 OF 08 | PROJECT DEVELOPMENT PHASE

## Project overview

LearnMate helps students work with scattered course materials through a REST API for uploads and AI-generated learning resources. The brief describes secure accounts, MongoDB storage, Gemini-powered content generation, and a dashboard-facing backend for saved study resources.

## Project description

A study-support API for uploading learning materials and generating summaries, flashcards, quizzes, and personalized study plans.

## Project status

Not stated in the source brief. Update this document when the team records an agreed status.

## Project capabilities

- Study material uploads
- AI summaries and flashcards
- Practice quizzes
- Personalized study plans

## Phase description

Build the solution, keep its code organized, and track which planned features are implemented.

## Project-specific phase focus

Implement secure material handling and the summary, flashcard, quiz, and plan workflows.

## Phase status and progress

Status: Not stated in the source material.
Progress: Not recorded in the source material.

## Tasks and requirements

[ ] Organize code into readable, reusable modules
[ ] Implement the core solution against requirements
[ ] Record feature names, modules, and implementation status
[ ] Review code layout and maintainability

## Relevant documents and data

Project brief: LearnMate - AI assistant for smarter learning and study support..docx

Phase templates:
- Code Layout, Readability and Reusability (5. Project Development Phase/Code-Layout, Readability and Reusability.pdf)

- Coding & Solution (5. Project Development Phase/Coding & Solution.pdf)
- Functional Features Included (5. Project Development Phase/No. of Functional Features Included in the Solution.pdf)

## Project technology stack

- Node.js
- Express.js
- MongoDB
- Mongoose
- Gemini AI

## Phase notes and decisions